

A-059

Roll No. 058

Total No. of Pages: 04

21N507 /

B.Tech. II Sem Main/Back (New Scheme) Acad. Session 2023-24

All Branches

(2FY2-07) – Basic Electrical Engineering

Time : 3 Hours

Maximum Marks: 70

Min. Passing Marks:

Instructions to Candidates:

Part – A: Short answer questions (up to 25 words) 10x2 marks = 20 marks. All 10 questions are compulsory.

Part – B: Analytical/Problem Solving questions 5x4 marks = 20 marks. Candidates have to answer 5 questions out of 7.

Part – C: Descriptive/Analytical/Problem Solving questions 3x10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1 NIL

2 NIL

PART-A

- Q.1. State and explain the Kirchhoff's voltage and current law. [2]
- Q.2. Write down different types of energy sources. [2]
- Q.3. Draw leading and lagging current phasor diagram for $V = V_{\max} \sin(\omega t)$ by an angle θ and β respectively. [2]
- Q.4. Write and discuss the transformer EMF equation. [2]
- Q.5. Explain the concept of voltage and current source transformation with an example. [2]
- Q.6. Explain commutator working in DC Motor. [2]
- Q.7. Draw the V-I Characteristics of p-n junction diode and mention the cut in voltage and peak inverse voltage. [2]
- Q.8. Explain statically and dynamically induced EMF with examples. [2]
- Q.9. What is Silicon Controlled Rectifier (SCR)? [2]
- Q.10. What is modulation in communication system? [2]

PART-B

- Q.1. Find the voltage V_{xy} in the network shown in Figure 1. [4]

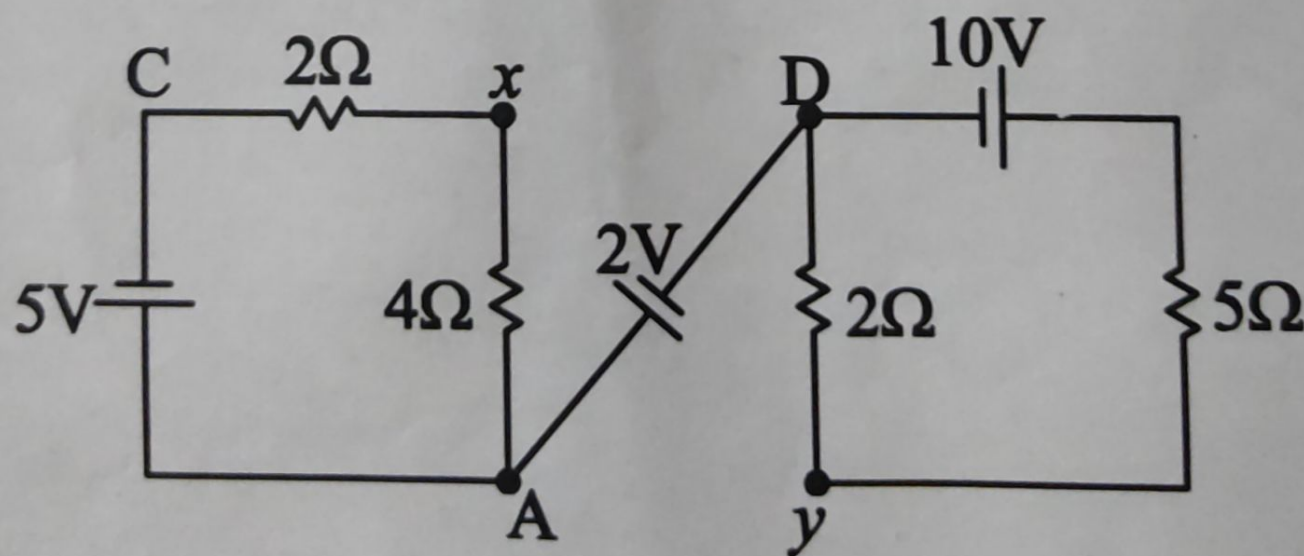


Figure 1

- Q.2. Find the current through $3\ \Omega$ resistor in the circuit shown in Figure 2, by using Thevenin's theorem. [4]

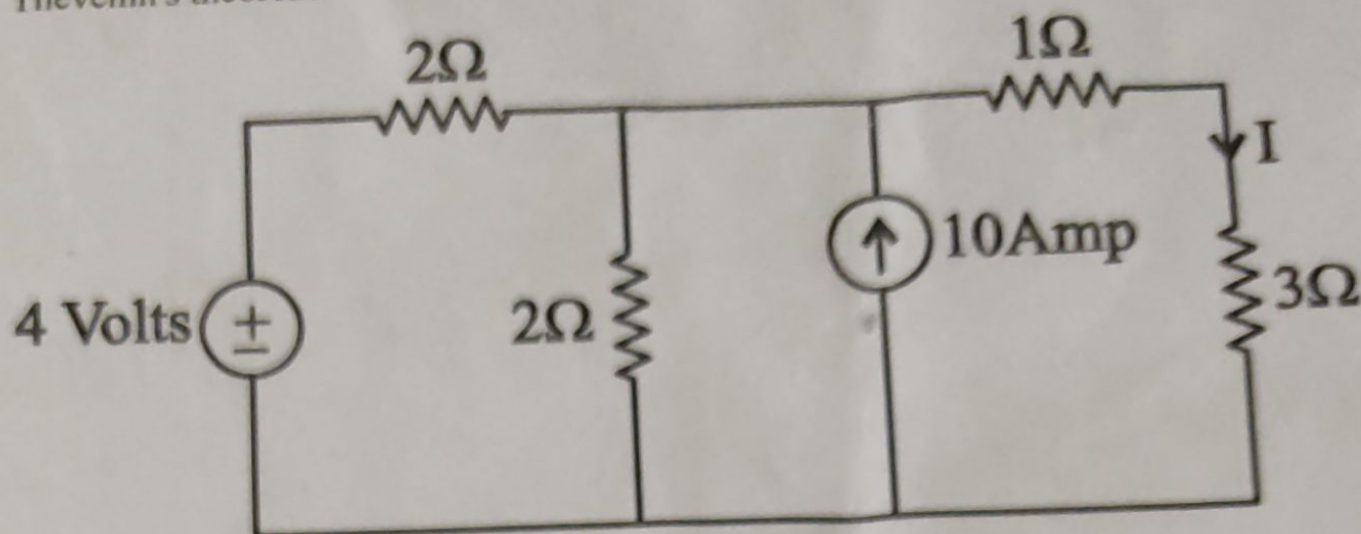


Figure 2

- Q.3. What are the construction and working principle of 1-phase transformer? Explain with suitable diagram. [4]
- Q.4. Briefly discuss the types of DC motors. What is the difference between differential compound and cumulative compound DC motor? [4]
- Q.5. What is an SCR? Sketch static V-I characteristics of a thyristor. Label the various voltages, currents and operating modes on this sketch. [4]
- Q.6. Discuss the amplitude modulation in detail. Derive the voltage equation of amplitude modulation. [4]
- Q.7. Explain the operation and V-I characteristics of Uni-Junction Transistor (UJT). Explain with suitable diagram. [4]

PART- C

- Q.1. Explain two wattmeter method of measuring power of 3-phase star connected load with the help of necessary phasor diagram and circuit diagram. [10]

- Q.2. Using nodal analysis, find the voltage across $12\ \Omega$ resistance in the circuit shown in Figure 3. [10]

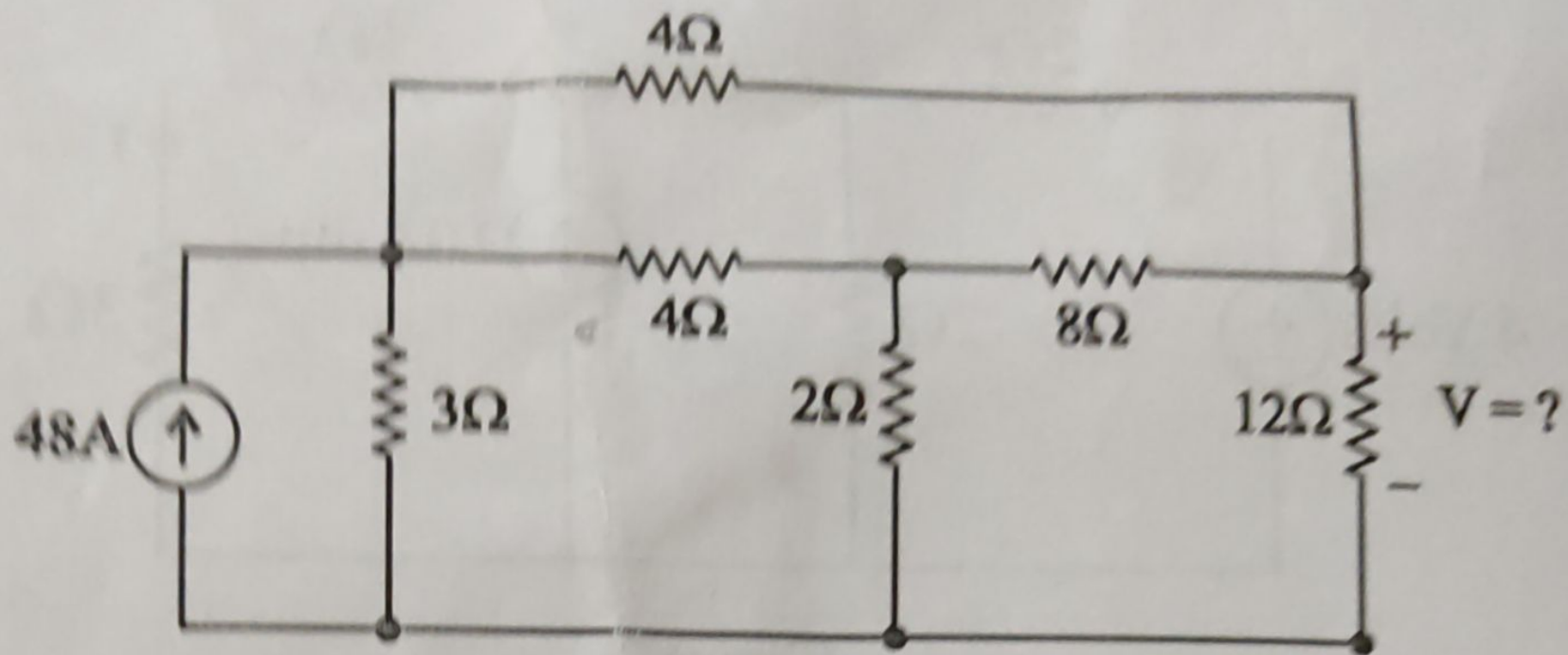


Figure 3

- Q.3. If an AC power supply of 100 V , 50 Hz is connected across a load of impedance $4+j8\ \Omega$. Calculate the current flowing through the circuit, active power, apparent power, reactive power and power factor. [10]
- Q.4. Draw the input and output characteristics of common emitter configuration and explain active saturation and cut-off region. [10]
- Q.5. Describe the different types of communications in details. [10]
